(a)

pydemi 0.2.0

INPUT

CHGCAR | Structure (ML)

MgO_225..._CHGCAR.gz
1.1MB

SUMMARY

Formula	Mg4O4
Atoms	8
Volume (Å³)	76.961
Electrons	32.000
Grid	64 × 64 × 64
Grid points	262,144
a, b, c (Å)	4.2536, 4.2536, 4.2536
α, β, γ (°)	90.00, 90.00, 90.00
Density	DFT (CHGCAR)

SETTINGS

Atomic partition

Nearest

HELP

Descriptor reference (PDF)

pydemi 0.2.0 · Shubham Maurya, CMS Lab, IIT Kanpur. Predicted densities: ChargE3Net, Koker et al., npj Comput. Mater. 2024.

Deploy

Mg4O4 MgO_225_DFT_CHGCAR.gz DFT density

Structure | Descriptors | Plots | Properties

All 56 | Bonding 19 | Structural 19 | Heterogeneity 18

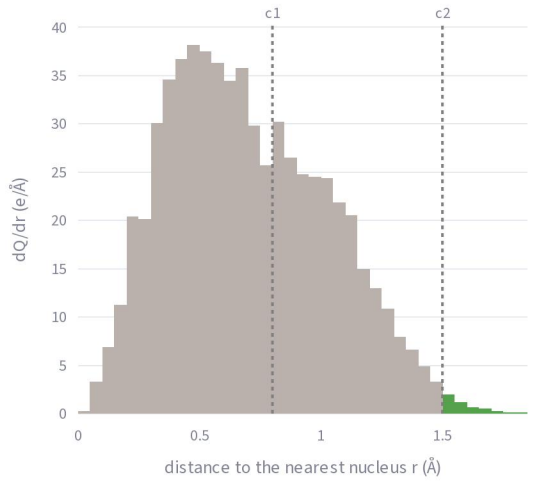
Search name or formula

Export

	descriptor	domain	value	units	fallback
☐	zeta	bonding	0.0242256	dimensionless	☐
☐	m1	bonding	0.703415	Å	☐
☐	m2	bonding	0.600198	Å²	☐
☐	sigma_r2	bonding	0.105405	Å²	☐
☐	f_core	bonding	0.626779	dimensionless	☐
☐	f_bond	bonding	0.366118	dimensionless	☐
☑	f_int	bonding	0.00710331	dimensionless	☐
☐	lnf	bonding	0.0352173	dimensionless	☐
☐	lnf_charge_weighted	bonding	0.364028	dimensionless	☐
☐	lap_concentration	bonding	0.5	dimensionless	☐
☐	lap_concentration_valence	bonding	0	dimensionless	☐
☐	ellip_bond_bounded_avg	bonding	0.247065	dimensionless	☐
☐	f_NCI	bonding	0.164261	dimensionless	☐
☐	NCI_attractive	bonding	0.384673	dimensionless	☐
☐	sign_lambda2_rho_mean	bonding	0.00111573	e/bohr³	☐

56 of 56 shown

f_int = 0.007103



Descriptor	f_int
Domain	bonding
Value	0.00710331
Units	dimensionless

FORMULA

$$f_{\text{int}} = \frac{\sum_{k:\, r_k > c_2} \rho_k}{\sum_k \rho_k}$$

Shells around the nearest nucleus: core $r_k \le c_1$, bond $c_1 < r_k \le c_2$,

(b)

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Deploy

Mg4O4 MgO_225_DFT_CHGCAR.gz DFT density

Structure | Descriptors | Plots | Properties

All 22 | Mechanical 4 | Elastic tensor 4 | Thermal 5 | Energetics 2 | Electronic 3 | Dielectric 4

Export

	property	group	prediction	DFT	± error	units	test score
☑	Bulk modulus	Mechanical	148.8	151.4	± 7.5	GPa	R^2 0.96
☐	Shear modulus	Mechanical	91.1	118.8	± 7.2	GPa	R^2 0.93
☐	Young's modulus	Mechanical	230	283	± 16	GPa	R^2 0.94
☐	Poisson ratio	Mechanical	0.224	0.189	± 0.023		R^2 0.65
☐	Universal anisotropy	Elastic tensor	0.32	0.22	± 0.79		R^2 0.30
☐	Elastic constant C11	Elastic tensor	294	274	± 20	GPa	R^2 0.94
☐	Elastic constant C12	Elastic tensor	80	90	± 11	GPa	R^2 0.89
☐	Elastic constant C44	Elastic tensor	109.9	141.0	± 9.4	GPa	R^2 0.89
☐	Debye temperature	Thermal	914	902	± 26	K	R^2 0.94
☐	Sound velocity, transverse	Thermal	5,790	5,844	± 186	m/s	R^2 0.93
☐	Sound velocity, longitudinal	Thermal	9,579	9,438	± 213	m/s	R^2 0.96
☐	Min. thermal conductivity, Cla	Thermal	2.518	2.393	± 0.060	W/(m·K)	R^2 0.96
☐	Min. thermal conductivity, Ca	Thermal	2.607	2.600	± 0.060	W/(m·K)	R^2 0.96
☐	Formation energy	Energetics	-2.900	-3.038	± 0.061	eV/atom	R^2 0.99
☐	Cohesive energy	Energetics	-4.90	-5.22	± 0.11	eV/atom	R^2 0.99
☐	Band gap (PBE)	Electronic	4.21	4.43	± 0.15	eV	R^2 0.93
☐	Metallic	Electronic	Not metal (> 99 %)	Not metal			AUC 0.99
☐	Direct gap	Electronic	Indirect (65 %)	Direct			AUC 0.77
☐	Dielectric constant, total	Dielectric	13.2	10.8	± 5.7		R^2 0.47
☐	Dielectric constant, electron	Dielectric	6.0	3.2	± 2.4		R^2 0.55

Property	Bulk modulus
Symbol	K
Prediction	148.8 GPa
DFT (mp-1265)	151.4 GPa
Typical error	± 7.5 GPa (test MAE)
Median error	4.2 GPa
Test R^2	0.961
R^2, 5 seeds	0.961 (spread < 0.001)
Inputs	descriptors + composition (193)
Trained on	4,847 materials
Tested on	606 materials
Model	v1 (tuned LightGBM, descriptors and composition)

Test set: 606 materials, MAE 7.5 GPa, R^2 0.961

